KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5053C | 5 Credits: 4 | Program Elective | Source-grounded

Two and Three Wheeler

 To identify different areas of two and three wheeler technology.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5053C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5053C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	5053C
Course title	Two and Three Wheeler
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

24 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To identify different areas of two and three wheeler technology.
- To find the applications of all the areas in day to day life.
- To appreciate the working and construction of different engines/ components.
- To understand the importance of electric two and three wheelers.

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding and frames of two wheelers. Illustrate the principle, construction and	See official syllabus
CO2	working of different types of engines and fuel 15 Understanding systems Explain the functions, construction and working	See official syllabus
CO3	15 Understanding of suspension system. Identify the features of three wheelers and electric	See official syllabus
CO4	14 Applying auto rickshaw	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 2 2
CO3	CO3 3 2
CO4	CO4 3 2 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	wheelers	4	As specified
Module 2	engines and fuel systems	7	As specified
Module 3	of suspension system.	5	As specified
Module 4	Identify the features of three wheelers and electric auto rickshaw	8	As specified

MODULE 1

wheelers

This module is presented from the official syllabus scope for Two and Three Wheeler. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: wheelers
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame

Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame is an official topic in Module 1 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the classification of two wheelers 3 understanding, illustrate the constructional details of frame should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3

3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3 is an official topic in Module 1 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding, and body know the wheels and tyres used in 2 and 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3 □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding, wheelers Summarize the Constructional details of Lithium

3 Understanding, wheelers Summarize the Constructional details of Lithium is an official topic in Module 1 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding, wheelers Summarize the Constructional details of Lithium using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding, wheelers summarize the constructional details of lithium should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding, wheelers Summarize the Constructional details of Lithium means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 3 Understanding, wheelers Summarize the Constructional details of Lithium [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of

5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of is an official topic in Module 1 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron

battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding, iron battery classification & layouts of two wheelers (motorcycles, scooters, mopeds). layout of mopeds, scooter and motor cycle. load on the frame, components of frame, mounting provisions of frame, tubular frame. engine based frame, twin spar frame, monocoque frame, frame material, body work. ergonomic considerations. spoked wheel, pressed steel wheels, alloy wheels. requirements of tyres, designation of tyres. cross ply and radial ply tyres, tyre with tube and tubeless tyres. inflation pressure. working principle, construction, charging and discharging, capacity rating of lithium-iron battery, battery management system of lead acid battery and lithium iron battery. illustrate the principle, construction and working of different types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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wheelers

  M1.01     Understand the classification of two wheelers             3
Understanding,

            Illustrate the constructional details of Frame
  M1.02                                                               3
Understanding,
            And Body
            Know the Wheels And Tyres used in 2 and 3
  M1.03                                                               3
Understanding,
            wheelers
            Summarize the Constructional details of Lithium
  M1.04                                                               5
Understanding,
            iron battery

Contents:
Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout
of
mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting
provisions of frame, Tubular frame. Engine based frame, Twin spar frame,
Monocoque
frame, Frame material, Body work.
Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels.
Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

engines and fuel systems

This module is presented from the official syllabus scope for Two and Three Wheeler. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: engines and fuel systems
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing

2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding, and four stroke engines. illustrate the valve timing and port timing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding, diagram,

2 Understanding, diagram, is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding, diagram, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding, diagram, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding, diagram, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Understanding, diagram, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the scavenging in two stroke engines 2 Understanding,

Explain the scavenging in two stroke engines 2 Understanding, is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the scavenging in two stroke engines 2 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the scavenging in two stroke engines 2 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the scavenging in two stroke engines 2 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ Explain the scavenging in two stroke engines 2 Understanding, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Outline the Lubrication and cooling 2 Understanding,

Outline the Lubrication and cooling 2 Understanding, is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the Lubrication and cooling 2 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the lubrication and cooling 2 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the Lubrication and cooling 2 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline the Lubrication and cooling 2 Understanding, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Summarize the Cranking System 2 Understanding,

Summarize the Cranking System 2 Understanding, is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the Cranking System 2 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the cranking system 2 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the Cranking System 2 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Summarize the Cranking System 2 Understanding, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Explain about the Ignition system 3 Understanding,

Explain about the Ignition system 3 Understanding, is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain about the Ignition system 3 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain about the ignition system 3 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain about the Ignition system 3 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain about the Ignition system 3 Understanding, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand the Electric Two Wheelers 2 Understanding, Working principle of two stroke and four stroke engines. Recent developments in engine. Fuel used – Gasoline, CNG, Diesel AND electric power plant. Types of scavenging and relative merits and demerits with scavenging systems. Systems requirements for Engine lubrication and cooling. Basic cranking mechanism (Roller type ratchet, Lock pawl type ratchet and regular ratchet wheel). Kick start mechanism. Lay out of kick start mechanism (Transmission kick start layout, primary kick start layout) . Auto start mechanism. Conventional ignition system- battery and magneto. Electronic Ignition System – Capacitor discharge ignition, ECU and various sensors used. Drivetrain layout of electric two wheeler Batteries, Electric motor, Motor controller, Charger and charging. Merits and demerits of electric two wheelers, High performance of electric wheelers Explain the functions, construction and working

Understand the Electric Two Wheelers 2 Understanding, Working principle of two stroke and four stroke engines. Recent developments in engine. Fuel used – Gasoline, CNG, Diesel AND electric power plant. Types of scavenging and relative merits and demerits with scavenging systems. Systems requirements for Engine lubrication and cooling. Basic cranking mechanism (Roller type ratchet, Lock pawl type ratchet and regular ratchet wheel). Kick start mechanism. Lay out of kick start mechanism (Transmission kick start layout, primary kick start layout) . Auto start mechanism. Conventional ignition system- battery and magneto. Electronic Ignition System – Capacitor discharge ignition, ECU and various sensors used. Drivetrain layout of electric two wheeler Batteries, Electric motor, Motor controller, Charger and charging. Merits and demerits of electric two wheelers, High performance of electric wheelers Explain the functions, construction and working is an official topic in Module 2 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the Electric Two Wheelers 2 Understanding, Working principle of two stroke and four stroke engines. Recent developments in engine. Fuel used – Gasoline, CNG, Diesel AND electric power plant. Types of scavenging and relative merits and demerits with scavenging systems. Systems requirements

for Engine lubrication and cooling. Basic cranking mechanism (Roller type ratchet, Lock pawl type ratchet and regular ratchet wheel). Kick start mechanism. Lay out of kick start mechanism (Transmission kick start layout, primary kick start layout) . Auto start mechanism. Conventional ignition system- battery and magneto. Electronic Ignition System – Capacitor discharge ignition, ECU and various sensors used. Drivetrain layout of electric two wheeler Batteries, Electric motor, Motor controller, Charger and charging. Merits and demerits of electric two wheelers, High performance of electric wheelers Explain the functions, construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the electric two wheelers 2 understanding, working principle of two stroke and four stroke engines. recent developments in engine. fuel used – gasoline, cng, diesel and electric power plant. types of scavenging and relative merits and demerits with scavenging systems. systems requirements for engine lubrication and cooling. basic cranking mechanism (roller type ratchet, lock pawl type ratchet and regular ratchet wheel). kick start mechanism. lay out of kick start mechanism (transmission kick start layout, primary kick start layout) . auto start mechanism. conventional ignition system- battery and magneto. electronic ignition system – capacitor discharge ignition, ecu and various sensors used. drivetrain layout of electric two wheeler batteries, electric motor, motor controller, charger and charging. merits and demerits of electric two wheelers, high performance of electric wheelers explain the functions, construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the Electric Two Wheelers 2 Understanding, Working principle of two stroke and four stroke engines. Recent developments in engine. Fuel used – Gasoline, CNG, Diesel AND electric power plant. Types of scavenging and relative merits and demerits with scavenging systems. Systems requirements for Engine lubrication and cooling. Basic cranking mechanism (Roller type ratchet, Lock pawl type ratchet and regular ratchet wheel). Kick start mechanism. Lay out of kick start mechanism (Transmission kick start layout, primary kick start layout) . Auto start mechanism. Conventional ignition system- battery and magneto. Electronic Ignition System – Capacitor discharge ignition, ECU and various sensors used. Drivetrain layout of electric two wheeler Batteries, Electric motor, Motor controller, Charger and charging. Merits and demerits of electric two wheelers, High performance of electric wheelers Explain the functions, construction and working means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Understand the Electric Two Wheelers 2 Understanding, Working principle of two stroke and four stroke engines. Recent developments in engine. Fuel used – Gasoline, CNG, Diesel AND electric power plant. Types of scavenging and relative merits and demerits with scavenging systems. Systems requirements for Engine lubrication and cooling. Basic cranking mechanism (Roller type ratchet, Lock pawl type ratchet and regular ratchet wheel). Kick start mechanism. Lay out of kick start mechanism (Transmission kick start layout, primary kick start layout) . Auto start mechanism. Conventional ignition system- battery and magneto. Electronic Ignition System – Capacitor discharge ignition, ECU and various sensors used. Drivetrain layout of electric two wheeler Batteries, Electric motor, Motor controller, Charger and charging. Merits and demerits of electric two wheelers, High performance of electric wheelers Explain the functions, construction and working □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
engines and fuel systems
            Understand the Working principle of two stroke
  M2.01                                                           2
Understanding,
            and four stroke engines.
            Illustrate the Valve timing and port timing
  M2.02                                                           2
Understanding,
            diagram,
  M2.03     Explain the scavenging in two stroke engines          2
Understanding,

  M2.04     Outline the Lubrication and cooling                   2
Understanding,

  M2.05     Summarize the Cranking System                         2
Understanding,

  M2.06     Explain about the Ignition system                     3
Understanding,

  M2.07     Understand the Electric Two Wheelers                  2
Understanding,

            Series Test – I                                       1

Contents:
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

of suspension system.

This module is presented from the official syllabus scope for Two and Three Wheeler. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of suspension system.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Illustrate the Steering geometry and effects 3 Understanding,

Illustrate the Steering geometry and effects 3 Understanding, is an official topic in Module 3 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the Steering geometry and effects 3 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the steering geometry and effects 3 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the Steering geometry and effects 3 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-[illegible] Illustrate the Steering geometry and effects 3 Understanding, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand the Suspension system 3 Understanding,

Understand the Suspension system 3 Understanding, is an official topic in Module 3 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the Suspension system 3 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the suspension system 3 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the Suspension system 3 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understand the Suspension system 3 Understanding, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain the Spring and shock absorber assembly 3 Understanding,

Explain the Spring and shock absorber assembly 3 Understanding, is an official topic in Module 3 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Spring and shock absorber assembly 3 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the spring and shock absorber assembly 3 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Spring and shock absorber assembly 3 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ Explain the Spring and shock absorber assembly 3 Understanding, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Outline the Telescopic suspension 3 Understanding,

Outline the Telescopic suspension 3 Understanding, is an official topic in Module 3 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the Telescopic suspension 3 Understanding, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the telescopic suspension 3 understanding, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the Telescopic suspension 3 Understanding, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline the Telescopic suspension 3 Understanding, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Explain the Transmission system 3 Understanding, Trail, Castor angle or Rack angle, wheel base. Steering column Construction Handle bar types and construction. Springer forks suspension, Girder forks suspension, Trailing and leading link suspension, telescopic fork suspension. Single link type front suspension, Double link type front suspension. Hardtail type rear suspension. Swing arm type rear suspension. Layout of two wheeler and three wheeler transmission system. Belt, chain and gear drive. Multi - plate clutch, centrifugal clutch. Assist slipper clutch. Gear box - Constant mesh gear box with ball lock mechanism. Sequential gear box. CVT - Continuous variable transmission Cush drive.

Explain the Transmission system 3 Understanding, Trail, Castor angle or Rack angle, wheel base. Steering column Construction Handle bar types and construction. Springer forks suspension, Girder forks suspension, Trailing and leading link suspension, telescopic fork suspension. Single link type front suspension, Double link type front suspension. Hardtail type rear suspension. Swing arm type rear suspension. Layout of two wheeler and three wheeler transmission system. Belt, chain and gear drive. Multi - plate clutch, centrifugal clutch. Assist slipper clutch. Gear box - Constant mesh gear box with ball lock mechanism. Sequential gear box. CVT - Continuous variable transmission Cush drive. is an official topic in Module 3 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Transmission system 3 Understanding, Trail, Castor angle or Rack angle, wheel base. Steering column Construction Handle bar types and construction. Springer forks suspension, Girder forks suspension, Trailing and leading link suspension, telescopic fork suspension. Single link type front suspension, Double link type front suspension. Hardtail type rear suspension. Swing arm type rear suspension. Layout of two wheeler and three wheeler transmission system. Belt, chain and gear drive. Multi - plate clutch, centrifugal clutch. Assist slipper clutch. Gear box - Constant mesh gear box with ball lock mechanism. Sequential gear box. CVT - Continuous variable transmission Cush drive. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the transmission system 3 understanding, trail, castor angle or rack angle, wheel base. steering column construction handle bar types and construction. springer forks suspension, girder forks suspension, trailing and leading link suspension, telescopic fork suspension. single link type front suspension, double link type front suspension. hardtail type rear suspension. swing arm type rear suspension. layout of two wheeler and three wheeler transmission system. belt, chain and gear drive. multi - plate clutch, centrifugal clutch. assist slipper clutch. gear box - constant mesh gear box with ball lock mechanism. sequential gear box. cvt - continuous variable transmission cush drive. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Transmission system 3 Understanding, Trail, Castor angle or Rack angle, wheel base. Steering column Construction Handle bar types and construction. Springer forks suspension, Girder forks suspension, Trailing and leading link suspension, telescopic fork suspension. Single link type front suspension, Double link type front suspension. Hardtail type rear suspension. Swing arm type rear suspension. Layout of two wheeler and three wheeler transmission system. Belt, chain and gear drive. Multi - plate clutch, centrifugal clutch. Assist slipper clutch. Gear box - Constant mesh gear box with ball lock mechanism. Sequential gear box. CVT - Continuous variable transmission Cush drive. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the Transmission system 3 Understanding, Trail, Castor angle or Rack angle, wheel base. Steering column Construction Handle bar types and construction. Springer forks suspension, Girder forks suspension, Trailing and leading link suspension, telescopic fork suspension.

Single link type front suspension, Double link type front suspension. Hardtail type rear suspension. Swing arm type rear suspension. Layout of two wheeler and three wheeler transmission system. Belt, chain and gear drive. Multi - plate clutch, centrifugal clutch. Assist slipper clutch. Gear box - Constant mesh gear box with ball lock mechanism. Sequential gear box. CVT - Continuous variable transmission Cush drive. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
of suspension system.

  M3.01     Illustrate the Steering geometry and effects          3
Understanding,

  M3.02     Understand the Suspension system                      3
Understanding,

  M3.03     Explain the Spring and shock absorber assembly        3
Understanding,

  M3.04     Outline the Telescopic suspension                     3
Understanding,

  M3.05     Explain the Transmission system                       3
Understanding,

Contents:
Trail, Castor angle or Rack angle, wheel base. Steering column Construction
Handle bar types and construction. Springer forks suspension, Girder forks
suspension,
Trailing and leading link suspension, telescopic fork suspension. Single link
type front
suspension, Double link type front suspension. Hardtail type rear suspension.
Swing arm
type rear suspension.
Layout of two wheeler and three wheeler transmission system. Belt, chain and gear
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Identify the features of three wheelers and electric auto rickshaw

This module is presented from the official syllabus scope for Two and Three Wheeler. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the features of three wheelers and electric auto rickshaw
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

1 Understanding wheelers

1 Understanding wheelers is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding wheelers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding wheelers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding wheelers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding wheelers [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the Layout of passenger auto rickshaw

Illustrate the Layout of passenger auto rickshaw is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the Layout of passenger auto rickshaw using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the layout of passenger auto rickshaw should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the Layout of passenger auto rickshaw means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the Layout of passenger auto rickshaw □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto

Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the layout of loading auto rickshaws 2 understanding make use of different types of loading auto should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Applying rickshaws

1 Applying rickshaws is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying rickshaws using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying rickshaws should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying rickshaws means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Applying rickshaws [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the Suspension system in three wheelers

Explain the Suspension system in three wheelers is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Suspension system in three wheelers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the suspension system in three wheelers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Suspension system in three wheelers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain the Suspension system in three wheelers □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Identify various brakes used in three wheelers 2 Applying Illustrate the working and construction of Frame

Identify various brakes used in three wheelers 2 Applying Illustrate the working and construction of Frame is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify various brakes used in three wheelers 2 Applying Illustrate the working and construction of Frame using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify various brakes used in three wheelers 2 applying illustrate the working and construction of frame should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify various brakes used in three wheelers 2 Applying Illustrate the working and construction of Frame means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Identify various brakes used in three wheelers 2 Applying Illustrate the working and construction of Frame □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

1 Understanding and body Summarize the construction and working of

1 Understanding and body Summarize the construction and working of is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and body Summarize the construction and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and body summarize the construction and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and body Summarize the construction and working of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 1 Understanding and body Summarize the construction and working of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding Electric Three Wheeler Drive train layout of passenger auto rickshaw, Drive train layout for loading auto rickshaw, Rear suspension system of passenger auto rickshaw. Swinging arm suspension, triangulation in swinging arm suspension. Rear suspension system of loading auto rickshaw, Layout of braking system hydraulic and mechanical of auto rickshaw. BLDC motor and AC induction motor construction and working. Cloud based mobility platform for vehicle tracking and monitoring (NEMO). Layout of electric auto rickshaw. Text / Reference: T/R Book Title/Author T1 Two and Three wheeler Technology - Dhruv U. Panchal R1 Irving. P. E., "Motor Cycle Engineering", Temple Press Book, London – 1992. R2 Newton Steed, "The Motor Vehicle", McGraw Hill Book Co. Ltd., New Delhi Two wheeler and three wheeler, Ramalingam K K SCITECH Publication, R3 Chennai R4 Siegfried Herrmann, "The Motor Vehicle", Asia Publishing House, Bombay. Online Resources: Sl.No Website Link 1 https://youtu.be/kTuybtMAiN8 2 https://youtu.be/DUK6gjbsLpQ 3 https://youtu.be/3E1SXG7VkQk

3 Understanding Electric Three Wheeler Drive train layout of passenger auto rickshaw, Drive train layout for loading auto rickshaw, Rear suspension system of passenger auto rickshaw. Swinging arm suspension, triangulation in swinging arm suspension. Rear suspension system of loading auto rickshaw, Layout of braking system hydraulic and mechanical of auto rickshaw. BLDC motor and AC induction motor construction and working. Cloud based mobility platform for vehicle tracking and monitoring (NEMO). Layout of electric auto rickshaw. Text / Reference: T/R Book Title/Author T1 Two and Three wheeler Technology - Dhruv U. Panchal R1 Irving. P. E., "Motor Cycle Engineering", Temple Press Book, London – 1992. R2 Newton Steed, "The Motor Vehicle", McGraw Hill Book Co. Ltd., New Delhi Two wheeler and three wheeler, Ramalingam K K SCITECH Publication, R3 Chennai R4 Siegfried Herrmann, "The Motor Vehicle", Asia Publishing House, Bombay. Online Resources: Sl.No Website Link 1 https://youtu.be/kTuybtMAiN8 2 https://youtu.be/DUK6gjbsLpQ 3 https://youtu.be/3E1SXG7VkQk is an official topic in Module 4 of Two and Three Wheeler. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Electric Three Wheeler Drive train layout of passenger auto rickshaw, Drive train layout for loading auto rickshaw, Rear suspension system of passenger auto rickshaw. Swinging arm suspension, triangulation in swinging arm suspension. Rear suspension system of loading auto rickshaw, Layout of braking system hydraulic and mechanical of auto rickshaw. BLDC motor and AC induction motor construction and working. Cloud based mobility platform for vehicle tracking and monitoring (NEMO). Layout of electric auto rickshaw. Text / Reference: T/R Book Title/Author T1 Two and Three wheeler Technology - Dhruv U. Panchal R1 Irving. P. E., “Motor Cycle Engineering”, Temple Press Book, London – 1992. R2 Newton Steed, “The Motor Vehicle”, McGraw Hill Book Co. Ltd., New Delhi Two wheeler and three wheeler, Ramalingam K K SCITECH Publication, R3 Chennai R4 Siegfried Herrmann, “The Motor Vehicle”, Asia Publishing House, Bombay. Online Resources: Sl.No Website Link 1 https://youtu.be/kTuybtMAiN8 2 https://youtu.be/DUK6gjbsLpQ 3 https://youtu.be/3E1SXG7VkQk using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding electric three wheeler drive train layout of passenger auto rickshaw, drive train layout for loading auto rickshaw, rear suspension system of passenger auto rickshaw. swinging arm suspension, triangulation in swinging arm suspension. rear suspension system of loading auto rickshaw, layout of braking system hydraulic and mechanical of auto rickshaw. bldc motor and ac induction motor construction and working. cloud based mobility platform for vehicle tracking and monitoring (nemo). layout of electric auto rickshaw. text / reference: t/r book title/author t1 two and three wheeler technology - dhruv u. panchal r1 irving. p. e., “motor cycle engineering”, temple press book, london – 1992. r2 newton steed, “the motor vehicle”, mcgraw hill book co. ltd., new delhi two wheeler and three wheeler, ramalingam k k scitech publication, r3 chennai r4 siegfried herrmann, “the motor vehicle”, asia publishing house, bombay. online resources: sl.no website link 1 https://youtu.be/ktuybtmain8 2 https://youtu.be/duk6gjbslpq 3 https://youtu.be/3e1sxg7vkqk should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Electric Three Wheeler Drive train layout of passenger auto rickshaw, Drive train layout for loading auto rickshaw, Rear suspension system of passenger auto rickshaw. Swinging arm suspension, triangulation in swinging arm suspension. Rear suspension system of

Aspect	What to prepare
	loading auto rickshaw, Layout of braking system hydraulic and mechanical of auto rickshaw. BLDC motor and AC induction motor construction and working. Cloud based mobility platform for vehicle tracking and monitoring (NEMO). Layout of electric auto rickshaw. Text / Reference: T/R Book Title/Author T1 Two and Three wheeler Technology - Dhruv U. Panchal R1 Irving. P. E., “Motor Cycle Engineering”, Temple Press Book, London – 1992. R2 Newton Steed, “The Motor Vehicle”, McGraw Hill Book Co. Ltd., New Delhi Two wheeler and three wheeler, Ramalingam K K SCITECH Publication, R3 Chennai R4 Siegfried Herrmann, “The Motor Vehicle”, Asia Publishing House, Bombay. Online Resources: Sl.No Website Link 1 https://youtu.be/kTuybtMAiN8 2 https://youtu.be/DUK6gjbsLpQ 3 https://youtu.be/3E1SXG7VkQk means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding Electric Three Wheeler Drive train layout of passenger auto rickshaw, Drive train layout for loading auto rickshaw, Rear suspension system of passenger auto rickshaw. Swinging arm suspension, triangulation in swinging arm suspension. Rear suspension system of loading auto rickshaw, Layout of braking system hydraulic and mechanical of auto rickshaw. BLDC motor and AC induction motor construction and working. Cloud based mobility platform for vehicle tracking and monitoring (NEMO). Layout of electric auto rickshaw. Text / Reference: T/R Book Title/Author T1 Two and Three wheeler Technology - Dhruv U. Panchal R1 Irving. P. E., “Motor Cycle Engineering”, Temple Press Book, London – 1992. R2 Newton Steed, “The Motor Vehicle”, McGraw Hill Book Co. Ltd., New Delhi Two wheeler and three wheeler, Ramalingam K K SCITECH Publication, R3 Chennai R4 Siegfried Herrmann, “The Motor Vehicle”, Asia Publishing House, Bombay. Online Resources: Sl.No Website Link 1 https://youtu.be/kTuybtMAiN8 2 https://youtu.be/DUK6gjbsLpQ 3 https://youtu.be/3E1SXG7VkQk □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Identify the features of three wheelers and electric auto rickshaw
            Outline the various Classification of three
  M4.01                                                             1
Understanding
            wheelers
  M4.02     Illustrate the Layout of passenger auto rickshaw        2
Understanding

  M4.03     Illustrate the Layout of loading auto rickshaws         2
Understanding
            Make use of different types of loading auto
  M4.04                                                             1
Applying
            rickshaws
  M4.06     Explain the Suspension system in three wheelers         2
Understanding

  M4.07     Identify various brakes used in three wheelers          2
Applying
            Illustrate the working and construction of Frame
  M4.08                                                             1
Understanding
            and body
            Summarize the construction and working of
  M4.09                                                             3
Understanding
            Electric Three Wheeler
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding, And Body Know the Wheels And Tyres used in 2 and 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding, wheelers Summarize the Constructional details of Lithium. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding, wheelers Summarize the Constructional details of Lithium*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding, iron battery Classification & layouts of two wheelers (motorcycles, scooters, mopeds). Layout of mopeds, scooter and motor cycle. Load on the frame, Components of frame, Mounting provisions of frame, Tubular frame. Engine based frame, Twin spar frame, Monocoque frame, Frame material, Body work. Ergonomic considerations. Spoked wheel, Pressed steel wheels, Alloy wheels. Requirements of tyres, Designation of tyres. Cross ply and radial ply tyres, Tyre with tube and tubeless tyres. Inflation pressure. Working principle, construction, charging and discharging, capacity rating of lithium-iron battery, Battery management system of lead acid battery and lithium iron battery. Illustrate the principle, construction and working of different types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding, diagram,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding, diagram,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain the scavenging in two stroke engines 2 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the scavenging in two stroke engines 2 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Outline the Lubrication and cooling 2 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the Lubrication and cooling 2 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain Illustrate the Steering geometry and effects 3 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the Steering geometry and effects 3 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Understand the Suspension system 3 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the Suspension system 3 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain the Spring and shock absorber assembly 3 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the Spring and shock absorber assembly 3 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Outline the Telescopic suspension 3 Understanding,. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the Telescopic suspension 3 Understanding,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain 1 Understanding wheelers. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding wheelers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the Layout of passenger auto rickshaw. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the Layout of passenger auto rickshaw*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the Layout of loading auto rickshaws 2 Understanding Make use of different types of loading auto*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Applying rickshaws. (3 marks)

Answer: Begin with the standard definition or identity of *1 Applying rickshaws*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Understand the classification of two wheelers 3 Understanding, Illustrate the constructional details of Frame**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding, and four stroke engines. Illustrate the Valve timing and port timing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the Steering geometry and effects 3 Understanding,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Understanding wheelers**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://youtu.be/kTuybtMAiN8 https://youtu.be/DUK6gjbsLpQ https://youtu.be/3E1SXG7VkQk

IN-PAGE SEARCH

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